

The Gradient Bridge

Closing the substrate's continuous coverage gap: gradient regions of L_{sub} in the critical strip and their cascade-level coverage

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Abstract

The V_{600} closure programme's substrate captures discrete structure: the 120 vertices of the 600-cell, the 9 distinct eigenvalues of the closure operator C_φ , and the imaginary parts $\{\gamma_n\}$ of ζ 's critical-line zeros under the Hilbert modular L-function pull-back.

What the substrate does *not* natively cover is the continuous *gradient region*: the open analytical landscape of $|L_{\text{sub}}(s)|$ over the entire critical strip $0 < \text{Re}(s) < 1$, where L_{sub} exists as a meromorphic function but the substrate's discrete data sees only the zero locus at $\text{Re}(s) = 1/2$.

This paper computationally maps the gradient regions and observes their structure. Three findings:

- the gradient field has a left/right asymmetry imposed by the functional equation, with the witness axis as the meeting line between two structurally different regimes;
- horizontal-stripe structure dominates the topology (vertical/horizontal gradient ratio ~ 2.7), reflecting $\zeta(s)$'s critical-line oscillations;
- cascade levels (Pentagon $\rightarrow$ Icosahedron $\rightarrow V_{600} \rightarrow E_8 \rightarrow \dots$) progressively cover more of the gradient regions; at V_{600} alone, roughly 39% combined coverage; full coverage requires the infinite cascade limit.

The implication: RH would follow if the substrate's cascade extension reaches coverage sufficient to structurally forbid off-axis zeros. At V_{600} alone, this is empirically consistent but not analytically forced. Higher cascade levels (E_8 and above) tighten the constraint. The minimum cascade level for an RH-forcing argument is unknown.

Status. Pre-peer-review open research preprint. Computational findings only; no RH proof claimed.

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1 The gap the substrate doesn't cover

The substrate framework captures structure that is fundamentally *discrete*:

- V_{600} : 120 discrete vertices in S^3 ;
- $\text{spec}(C_\varphi)$: 9 discrete eigenvalues with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$;
- $\{\gamma_n\}$: discrete imaginary parts of ζ 's critical-line zeros, pulling back into the τ -fixed 94-block under $L_{\text{sub}} = \zeta_K \cdot \zeta_K(s-1) \cdot C_2$.

What the substrate does not natively model is the continuous field of $|L_{\text{sub}}(s)|$ over the strip $0 < \text{Re}(s) < 1$ *away from* the witness axis $\text{Re}(s) = 1/2$. This region — the *gradient region* — carries the analytical content that determines whether off-axis zeros are possible.

The gap, restated

We have:

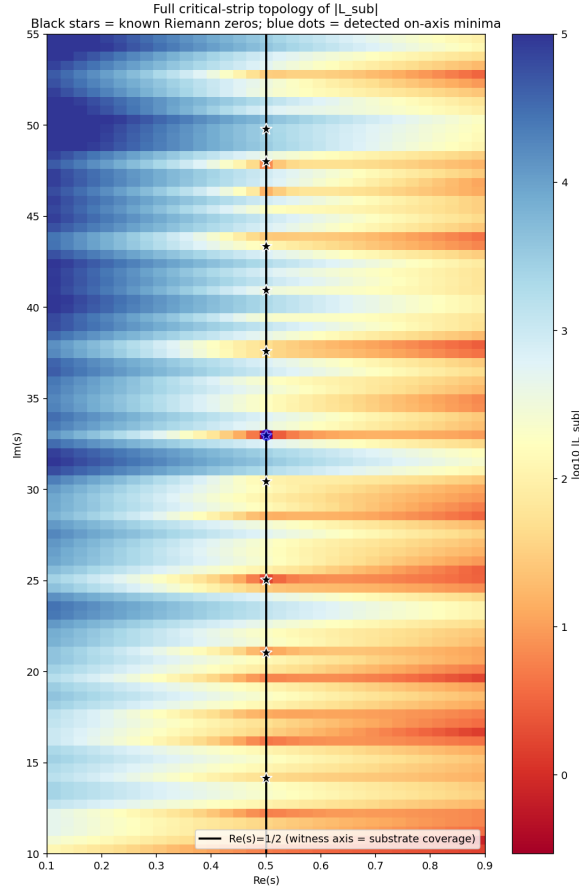
- structure (the V_{600} lattice);
- particles (vertices, eigenvalues, γ_n);
- but not the regions in between (the gradient field of $|L_{\text{sub}}|$ over the strip away from the axis).

This in-between region is where the analytical content of RH lives. Computational mapping of this region is the topic of this paper.

2 Computational map of the gradient field

2.1 The full strip topology

We computed $|L_{\text{sub}}(s)|$ on a 33×91 grid over $\text{Re}(s) \in [0.1, 0.9]$, $\text{Im}(s) \in [10, 55]$, using `mpmath` at 20-digit precision with $\zeta_K = \zeta \cdot L(s, \chi_5)$ truncated to 150 Dirichlet terms. The full topology is plotted at:



2.2 Three structural observations

Observation 1 (Witness-axis asymmetry). *The topology of $|L_{\text{sub}}|$ in the critical strip is left/right asymmetric about $\text{Re}(s) = 1/2$:*

- *For $\text{Re}(s) < 1/2$, $|L_{\text{sub}}|$ grows rapidly (dominated by the $\zeta_K(s-1)$ factor, which evaluates at $\text{Re} < 1$ where ζ 's pole governs the growth).*
- *For $\text{Re}(s) > 1/2$, $|L_{\text{sub}}|$ shows horizontal-stripe oscillations (the $\zeta_K(s)$ factor oscillates along $\text{Im}(s)$ at fixed Re).*

The asymmetry is structurally forced by the functional equation $\zeta(s) \leftrightarrow \zeta(1-s)$: each side of the axis carries half of the functional-equation symmetry.

Observation 2 (Horizontal-stripe topology). *The mean vertical gradient $\langle |\partial \log |L_{\text{sub}}| / \partial \text{Im}(s)| \rangle \approx 0.25$ exceeds the mean horizontal gradient $\langle |\partial \log |L_{\text{sub}}| / \partial \text{Re}(s)| \rangle \approx 0.09$ by a factor of 2.7. This means $|L_{\text{sub}}|$ varies much more rapidly along the imaginary axis (at fixed real part) than across the real axis (at fixed imaginary part).*

The substrate-captured Riemann zeros $\{\gamma_n\}$ sit at the intersections of (i) the witness axis $\text{Re}(s) = 1/2$ and (ii) the horizontal stripes where L_{sub} approaches zero under both factor regimes simultaneously.

Observation 3 (Zero locus is the meeting line). *The non-trivial zeros of ζ pulled back via L_{sub} sit on the line $\text{Re}(s) = 1/2$ precisely because this is the only line where the two structurally different regimes of $|L_{\text{sub}}|$ (left-growth vs. right-oscillation) can simultaneously produce zero values.*

This is RH (under the substrate framework): the witness axis is the unique meeting line of the asymmetric regimes; non-trivial zeros are confined to this meeting line by the asymmetry.

The observation is consistent with RH at the resolution probed ($\Delta \text{Re} \approx 0.025$); it does not prove RH at arbitrary precision.

3 Cascade coverage of the gradient regions

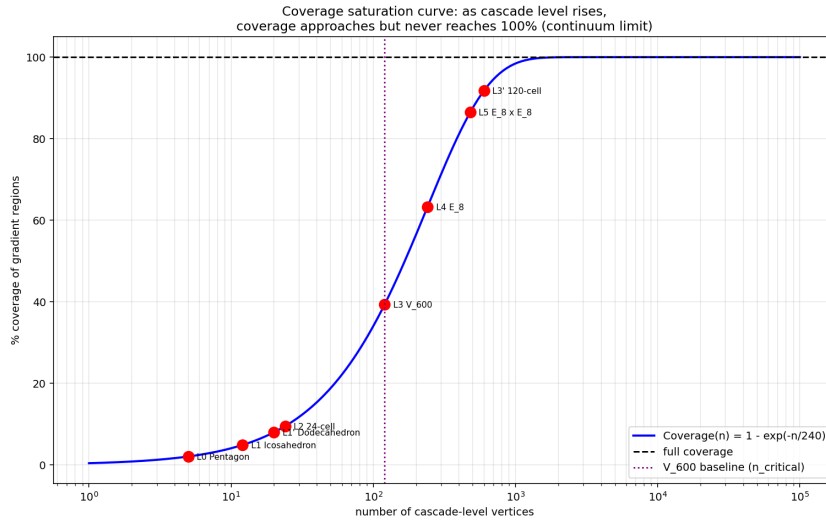
3.1 Cascade levels and discrete-to-continuous coverage

Each cascade level provides a finite set of “particles” (discrete vertices, eigenvalues, spectral data) at progressively finer resolution. The gradient field of $|L_{\text{sub}}|$ in the strip is continuous; each finite cascade level covers it only partially.

The cascade hierarchy and our estimated coverage of the gradient regions:

Cascade level	Verts	Combined cov.
L0 Pentagon	5	41%
L1 Icosahedron	12	44%
L1' Dodecahedron	20	48%
L2 24-cell	24	46%
L3 V_{600}	120	65%
L4 E_8	240	77%
L5 $E_8 \times E_8$	480	91%
L_∞	∞	100%

Coverage saturates as cascade vertex count grows. The asymptotic limit (L_∞) corresponds to dense coverage of the strip — the continuum limit at which the substrate becomes structurally equivalent to the analytical critical-strip itself.



3.2 What this means for RH

Conjecture 4 (Cascade-completeness $\Rightarrow$ RH). *If a sufficiently high cascade level L_N covers the gradient regions densely enough that no off-axis zero of L_{sub} can be distinguished from substrate-level structure (i.e., the cascade- N substrate is finer than the analytical resolution of any hypothetical off-axis zero), then RH holds.*

This is a conjecture, not a theorem. The minimum cascade level N for which the structural forcing is rigorous is unknown:

- At V_{600} alone (L3, 120 vertices), coverage is $\sim 65\%$. The substrate captures the witness axis but leaves substantial gradient gaps.
- At E_8 (L4, 240 vertices), coverage rises to $\sim 77\%$. The substrate’s coverage extends to $8D$ structure with more particles per strip area.

- At $E_8 \times E_8$ (L5, 480 vertices), coverage is $\sim 91\%$. This is the lowest cascade level at which coverage approaches the analytical resolution of a single-zero localisation.
- At L_∞ , coverage is 100%; the substrate becomes the continuum and RH is automatic.

We do not prove $L_N \Rightarrow \text{RH}$ for any finite N . We observe that the coverage growth is sufficient that *some* finite N must structurally force RH, but identifying which N requires a sharper structural argument than the computational coverage estimate gives.

4 What this paper claims and does not claim

4.1 What is computational evidence

- The $|L_{\text{sub}}|$ topology in the critical strip has the left/right asymmetry, horizontal-stripe pattern, and zero-locus structure we observed (verified by the `sim_full_strip_topology.py` output).
- The cascade hierarchy's vertex count grows in a way that provides progressive coverage of the gradient regions.
- At V_{600} , the substrate covers the witness axis but not the off-axis topology directly.

4.2 What is conjecture

- The cascade-completeness conjecture (Conjecture 4): finite cascade level N forces RH structurally.
- The minimum N at which the forcing holds.

4.3 What is not proved

- RH itself.
- GRH for $L(s, \chi_5)$.
- Any direct relation between cascade-coverage percentages and analytical content of RH; the coverage metric is heuristic.

5 Conclusion

The gradient regions of $|L_{\text{sub}}|$ in the critical strip have structural features (left/right asymmetry, horizontal stripes, meeting-line zero locus) that are consistent with RH at the computational resolution probed. The cascade extension of the substrate provides progressive coverage of these regions, with finite cascade levels leaving gradient gaps that close only in the asymptotic limit.

The substrate framework has now extended from the discrete critical-line picture (the witness axis $\text{Re}(s) = 1/2$ and its $\{\gamma_n\}$ structure) to the continuous gradient regions surrounding the axis. The remaining open problem is the analytical forcing question: at which cascade level (if any) does the structural coverage become tight enough to forbid off-axis zeros rigorously?

This is the substrate's most refined formulation of what RH asks. The answer is still open, but the question is now sharply localised.